

Spark Local Installation on Linux

1. Update/Upgrade system

```
cd ~  
sudo apt update && sudo apt upgrade
```

2. Install jdk/java.

```
sudo apt install jdk-21-jre-headless
```

3. Install Python with miniconda and do necessary setup.

```
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh`  
bash ~/Miniconda3-latest-Linux-x86_64.sh`  
rm -rf ~/Miniconda3-latest-Linux-x86_64.sh
```

4. Install Apache Spark.

```
wget https://d1cdn.apache.org/spark/spark-4.1.1/spark-4.1.1-bin-hadoop3.tgz  
tar -xzf spark-4.1.1-bin-hadoop3.tgz  
mkdir spark  
mv spark-4.1.1-bin-hadoop3 spark/spark-4.1.1-bin-hadoop3  
rm -rf spark-4.1.1-bin-hadoop3.tgz
```

5. Add environment variables for spark/pyspark setup.

```
export PYSPARK_PYTHON="/root/miniconda3/bin/python"  
export SPARK_HOME="/root/spark/spark-4.1.1-bin-hadoop3"  
export PYTHONPATH="/root/spark/spark-4.1.1-bin-hadoop3/python:/root/spark/spark-4.1.1-bin-hadoop3/python/lib/py4j-0.10.9.9-src  
source ~/.bashrc
```

6. Test run the pyspark

```
pyspark
```